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Analyzing Greenland's Geochemistry through Machine Learning

Research Project Report

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Abstract

Greenland is among the world’s most mineral-rich but least-explored regions, and the Geological Survey of Denmark and Greenland (GEUS) publishes decades of geochemical sampling through open web services. This project asks what machine learning can recover from that archive, and focuses on one question in depth: given a rock or sediment sample with some elements measured, can a held-out element be predicted, and does knowing *where* the sample came from help?

We build a single pipeline and run it over two independent GEUS datasets — whole-rock geochemistry (6,957 samples, clustered along the West Greenland margin) and stream sediment (17,463 samples, spanning the whole ice-free coast). A sample’s own chemistry predicts a held-out element well, reaching a mean R^2 of 0.868 on stream sediment and 0.798 on whole rock across 71 and 48 target elements respectively. Spatial context adds remarkably little on top of it: nearest-neighbour chemistry, three full-coverage geophysical grids, and raw coordinates each contribute less than one fold standard deviation, and are largely interchangeable with one another. A graph neural network with learned, distance-weighted aggregation does not beat a hand-specified average over the same neighbourhood.

The gain is not, however, uniform. Stratifying targets by how well own chemistry already predicts them shows a clean monotone decay — roughly $+0.05 R^2$ for elements chemistry predicts poorly, and nothing at all for elements it already predicts above 0.95 — and this curve reproduces almost exactly across the two datasets despite their different sampling geometries and target sets. The elements that benefit are a coherent group: the hydrothermal pathfinder suite (As, Sb, Pb, Cs on both datasets, plus U on stream sediment, which rises from 0.594 to 0.814). We also report a methodological result we think matters more than any single number: an earlier version of this analysis, scored on each dataset’s ten best-covered elements, appeared to show a strong contrast between the two sampling geometries. Widening the target set to a coverage threshold dissolved that contrast entirely, and 65 % of the apparent effect turned out to be a single element. Target selection, not model choice, was the first-order term.

All results use random k -fold cross-validation. Spatially blocked evaluation, which is where these conclusions are most likely to change, is set out as the next stage of the work.

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1 Introduction

Mineral exploration in Greenland has produced a large but fragmented public record. Since the 1950s, government surveys and exploration companies have sampled rock outcrops and stream catchments along the ice-free margin, and the resulting analyses — tens of thousands of geochemical measurements, of which 46 226 are used here — are published by the Geological Survey of Denmark and Greenland (GEUS) through open web services and the Greenland Mineral Resources Portal [1].

The archive is attractive to machine learning for the same reasons it is awkward. Coverage is heterogeneous: some outcrops carry dozens of samples while entire regions carry none. Analytical suites differ by campaign, so the element matrix is sparse and its sparsity is not random. Measurements are compositional and span six orders of magnitude, from weight percent to parts per billion. And everything is spatially autocorrelated, because neighbouring catchments drain the same lithologies [2].

1.1 Research questions

Following the project agreement with the supervisor, the work was organised around three questions:

RQ0 — report parsing. Use a public large language model to convert scanned exploration reports into structured records. *Dropped by agreement with the supervisor:* the scanned reports rarely contain the assay tables that would make parsing useful downstream, and the geochemical database is already available in structured form.

RQ1 — missing geochemistry. Given a sample with some elements measured, predict a held-out element from (a) the sample’s other elements and (b) its spatial neighbourhood. *This report.*

RQ2 — unexplored regions. Predict concentrations in a geological area held out *spatially* from training, rather than at random. *Scoped and prepared; see Chapter 6.*

RQ2 as stated was initially ambiguous — “predict areas where no reports are available” could mean predicting concentrations on unsampled ground, or predicting which regions are prospective for mineralisation. The supervisor’s clarification settles it as the first: selecting a geological area outside the one used for training and predicting the numbers there. This is a regression problem under spatial hold-out, not occurrence classification.

1.2 Contributions

1. A single preprocessing and evaluation pipeline applied unchanged to *two* independent GEUS datasets with markedly different sampling geometries, allowing every claim to be checked for dataset-specificity rather than asserted from one.
2. A systematic comparison of four ways of encoding a spatial neighbourhood as tabular features, across seven neighbourhood sizes. The result is unambiguous and holds on both datasets and under two model families: a neighbourhood should be *pooled*, not *enumerated*.
3. Evidence that spatial context — whether supplied as neighbouring chemistry, geophysical grids, or raw coordinates — is largely redundant with a sample’s own chemistry, and that the residual value is concentrated in a geologically coherent set of mobile and pathfinder elements.
4. A graph neural network evaluated against the hand-built encodings on identical folds, including the specific question of whether edge distance carries information that fixed-radius averaging cannot. It does not.
5. A methodological caution (Section 4.5): scoring on the n best-covered elements, a common convenience choice, manufactured an apparent effect that vanished under a principled coverage threshold.

All experiments are reproducible from three notebooks against files committed to the repository; every reported number is cached as JSON alongside the code.

2 Background

2.1 Geochemical data

Geochemical analyses are *compositional*: concentrations are parts of a whole and are constrained to sum to a constant, which makes their covariance structure inherently negative and their raw values ill-suited to methods assuming unbounded, symmetric errors [3]. The practical consequences for this work are two. First, concentrations spanning weight percent to parts per billion are log-transformed before modelling, which is standard practice and makes errors comparable across elements of very different abundance. Second, elements are strongly co-determined — a sample rich in SiO_2 is necessarily poor in something else — which is precisely why predicting a held-out element from its companions is feasible at all, and why it sets a demanding baseline for any spatial method to beat.

A second feature of survey geochemistry is the *detection limit*. An analysis that finds nothing reports not zero but “below detection”, often encoded as a negative or sentinel value. Treating such entries as numbers rather than as missing data biases every downstream statistic, so they are masked before any aggregation (Section 3.2).

2.2 Spatial structure and its evaluation

Geochemical samples are spatially autocorrelated: nearby samples resemble each other because they derive from the same rocks [2]. This is simultaneously the motivation for spatial modelling — a neighbour’s chemistry is informative — and the principal threat to its evaluation. Under random cross-validation, a test sample frequently has training samples a few hundred metres away, so a model can appear to generalise spatially when it is in fact interpolating between near-duplicates. Roberts et al. [4] survey this failure mode and recommend blocking folds in space so that test and training data are separated by a distance comparable to the autocorrelation range.

This report uses random k -fold cross-validation throughout, and its results should be read accordingly. Spatial blocking is the subject of RQ2 and is deliberately not conflated with the present results; Section 5.5 states precisely which conclusions are expected to be sensitive to it. Two design choices partially mitigate leakage even here: neighbour features are constructed from training coordinates only, and co-located

samples can be excluded from a sample’s neighbourhood so that “another sample from the same outcrop” is never mistaken for genuine spatial generalisation.

2.3 Models for tabular and graph-structured data

Gradient-boosted decision trees remain the strongest general-purpose learners on tabular data of this size and heterogeneity, typically outperforming neural architectures without extensive tuning [5]. We use the histogram-based gradient boosting implementation in scikit-learn [6], which follows the binning strategy introduced by LightGBM [7]: features are discretised before splits are searched, which makes training near-linear in sample count and handles missing values natively — an important property here, since the element matrix is sparse by construction. Random forests [8] and XGBoost [9] are included as comparison points.

Graph neural networks offer a different way to use a neighbourhood: rather than summarising it into fixed columns, they learn an aggregation function over a graph. GRAPH-SAGE [10] aggregates neighbour representations with a permutation-invariant operator and is distance-blind by construction. Graph attention networks [11] instead learn per-edge weights, and admit edge features — which allows the network to learn how quickly a neighbour’s relevance decays with distance, something a fixed average over the nearest k samples cannot express. That contrast is the reason both architectures appear here. Implementations use PyTorch Geometric [12].

3 Methodology

3.1 Datasets

Two GEUS geochemistry datasets are used, both public and both processed by the same code. They were chosen because they differ sharply in sampling geometry, which lets every conclusion be tested for dataset-specificity (Table 3.1).

Table 3.1: The two datasets after preprocessing. The two differ in sampling geometry: whole rock is clustered exploration sampling in a narrow longitude band, while stream sediment covers the whole ice-free margin with a fifth of the co-location.

	Whole rock	Stream sediment
Raw rows	26 068	20 158
Raw element columns	76	523
Merged physical samples	6957	17 463
Element columns after merging	76	93
Elements passing coverage filter	68	87
Target elements (coverage ≥ 1000)	48	71
Unique coordinate pairs	4498	16 542
Samples sharing a coordinate	3458 (50 %)	1759 (10 %)
Maximum samples at one coordinate	67	11
Longitude span	−53.9 to −42.3 (11.7°)	−72.8 to −13.3 (59.5°)
Latitude span	59.9° to 70.2°	59.8° to 83.6°

Stream sediment’s 523 raw element columns are 93 elements each measured by up to four laboratory and method combinations (for instance ACTLABS FUS ICP Ni ppm alongside ACTLABS INAA Ni ppm); whole rock already carries one column per element. Co-location is genuine sampling density rather than duplication — 67 samples at a single outcrop reflects a detailed campaign — and is therefore retained in both datasets, with its effect measured separately (Section 3.4).

Each source file carries a schema in JSON naming the sample identifier, metadata, physical and location columns; element columns are whatever remains. Adding a dataset therefore requires a schema file and three constants, not a code change. Longitude and latitude are classified as metadata, so coordinates never enter the chemistry feature set by accident.

3.2 Preprocessing

The pipeline is six steps, and their order is load-bearing:

1. **Mask below-detection values.** Entries ≤ 0 encode “below detection limit” and become missing. This is done *first*, so that a negative reading can never contaminate a median computed later.
2. **Merge analytical columns.** Per-laboratory, per-method columns of the same element are collapsed into one by taking the median across whichever methods measured that sample. Methods differ in detection limit and bias, so the median is a deliberate choice of the neutral option rather than an obvious one. Stream sediment goes from 523 columns to 93; whole rock is untouched.
3. **Merge by sample number.** The median across duplicate rows describing one physical sample.
4. **Log-transform.** Base-10 logarithm, applied after both merges, so the result is a log of medians rather than a median of logs.
5. **Coverage filter.** Element columns measured in fewer than 10 samples are dropped. This is last on purpose: filtering earlier would discard measurements the two merges would have rescued.
6. **Splits.** An 80/10/10 holdout and a five-fold random cross-validation assignment, both written to disk so every experiment sees identical folds.

3.3 Target selection

An experiment of this shape requires a choice of which elements to predict. The obvious convenience choice — the n best-covered elements — turns out to be consequential, and Section 4.5 reports what went wrong with it. All headline results in this report use a *coverage threshold* instead: every element measured in at least 1000 samples qualifies as a target, giving 48 elements on whole rock and 71 on stream sediment. The threshold is principled in a way a top- n cut is not, because it does not depend on how many elements a dataset happens to contain.

3.4 Neighbour features

Neighbour features are constructed at training time, per fold, and never cached to disk. A k -d tree is fitted on *training coordinates only* and queried with all rows, so a test sample can never see another test sample. Self-matches are pushed to infinite distance. An optional flag additionally discards distance-zero neighbours, which separates “another

sample from the same outcrop” from “a genuinely different place”; all results below use it.

Four encodings of the resulting neighbourhood are compared. With k neighbours and m elements, they differ in how much they aggregate (Table 3.2).

Table 3.2: The four neighbour encodings. Column counts are for $k = 20$ and $m = 87$ (stream sediment). Only variant C has a width independent of k .

Variant	Content	Added columns
A	Mean of the <i>target</i> element over k neighbours	1
B	The k individual <i>target</i> values	20
C	Mean of <i>every</i> element over k neighbours	87
D	<i>Every</i> element of <i>every</i> neighbour	1740

The original row-wise construction of these features was replaced with vectorised NumPy gathers, verified to produce identical output. The rewrite reduced construction from 65.8 min to 0.077 s per fold, a factor of roughly 51 600, which is what made the seven-point k -sweep across two datasets affordable at all.

3.5 Geophysical features

Geochemical neighbours only exist where somebody has already sampled; geophysical grids are measured everywhere. They are therefore the natural substitute for a neighbourhood on unexplored ground, and their inclusion answers the supervisor’s request for a geologically motivated input rather than a purely geometric one.

Three grids were included, on the criterion of *complete Greenland coverage* assessed over the whole landmass rather than over the samples at hand: magnetic anomaly [13], geothermal heat flow [14], and depth to Moho [15]. All three are valid at 100 % of samples. Magnetic anomaly tracks how magnetite-rich the underlying rock is; heat flow reflects tectonic setting and the decay of U, Th and K; Moho depth is crustal thickness, thin under rifted margins and thick under old craton.

Several candidate products were rejected, and the reason is worth recording because it is not visible from the files themselves. North Atlantic gravity compilations contain no missing values over Greenland — but only because they contain no *measurement points* over western Greenland, at a median distance of 157 km from a sample to the nearest grid node. Absence of NaN is not evidence of coverage.

3.6 Models

Three tabular learners were compared on identical folds before fixing one for the remaining experiments: a random forest [8] with 200 trees, histogram-based gradient boosting [6] with 200 iterations, and XGBoost [9] with 200 estimators and the `hist` tree method. Results are in Section 4.2. Gradient boosting was fastest and most accurate and is used throughout; the random forest is retained as an independent check that conclusions are not an artefact of one inductive bias.

3.7 Graph neural network

The graph has one node per sample and directed edges from each of a sample’s k nearest *training* neighbours, using exactly the mapping the tabular encodings use, so the network sees the same neighbourhood the hand-built columns saw. Each edge carries $\log(1 + d)$ with d in kilometres as its feature.

Node features are the sample’s own elements with the target removed. Because a network cannot consume a missing value, each element contributes two channels: the standardised value, zeroed where absent, and a binary present/absent flag, so the model can distinguish “average” from “not measured”. Standardisation statistics come from training rows only.

Two architectures are compared, and the contrast is the experiment. GRAPH-SAGE [10] aggregates neighbours without reference to distance. A graph attention network [11] with `edge_dim=1` computes attention weights per edge *from the distance*, which is the one thing variants A–D structurally cannot encode. Both use two message-passing layers of hidden width 64, a skip connection carrying the node’s own features directly to a two-layer head, dropout 0.1, and Adam at learning rate 10^{-2} with weight decay 10^{-4} .

Training is full-batch and transductive: the whole graph is passed forward each epoch, but the loss only ever sees training nodes. A validation slice is carved out of the *training* nodes to select the stopping epoch, so test data is never inspected during fitting. Runs were executed on a Colab T4 GPU at approximately 9.1 s per fit at $k = 20$.

3.8 Evaluation

Every experiment reports the coefficient of determination R^2 on the log-transformed scale, per target element, averaged over five random folds; tables report the mean over target elements. Only samples where the target was actually measured are scored.

Because a mean over dozens of elements hides a great deal, we also report the standard deviation across folds, averaged over elements, as a noise floor. It is 0.0180 on whole rock and 0.0137 on stream sediment. Differences smaller than this are not interpreted as real, and are reported in units of it where the comparison matters.

4 Results

4.1 Own chemistry is a strong predictor

Predicting a held-out element from a sample’s remaining elements reaches a mean R^2 of 0.7983 on whole rock (48 targets) and 0.8676 on stream sediment (71 targets). This is a demanding baseline, and it is the number every spatial method in this report must improve upon.

The two datasets are not directly comparable on this figure, because their target sets differ: whole rock’s 48 elements include a higher proportion of trace metals, stream sediment’s 71 include a large suite of rare earth elements that are near-collinear with one another and consequently easy to predict. Comparisons in this chapter are therefore always *within* a dataset.

4.2 Model selection

On whole rock, gradient boosting outperformed both alternatives on every element while also being the fastest (Table 4.1). XGBoost at default settings was indistinguishable from the random forest, which is consistent with the observation that its competitive reputation rests on extensive tuning; its untuned defaults overfit a dataset of this size and correlation structure.

Table 4.1: Model comparison on whole rock, ten elements, five folds, own-chemistry features.

Model	Mean R^2	Total time
Random forest (200 trees)	0.8207	388 s
Gradient boosting (200 iterations)	0.8388	155 s
XGBoost (200 estimators, <code>hist</code>)	0.8215	199 s

4.3 The four neighbour encodings

Table 4.2 gives the four encodings on both datasets at $k = 20$, with co-located neighbours excluded. The shape is the same on both: variant C is best, variant D lands at or below the baseline, and every gain is smaller than one fold standard deviation.

Table 4.2: The four neighbour encodings, $k = 20$, co-located neighbours excluded, coverage-threshold target sets. Fold standard deviation is 0.0180 (whole rock) and 0.0137 (stream sediment).

Variant	Whole rock (48)		Stream sediment (71)	
	Mean R^2	Δ	Mean R^2	Δ
Baseline (no neighbours)	0.7983	—	0.8676	—
A	0.8043	+0.0059	0.8757	+0.0081
B	0.8050	+0.0066	0.8724	+0.0048
C	0.8132	+0.0149	0.8801	+0.0125
D	0.7982	−0.0001	0.8659	−0.0017

Expressed against the noise floor, variant C is worth 0.82σ on whole rock and 0.91σ on stream sediment. **On both datasets the best available neighbour encoding buys less than one fold standard deviation.**

4.4 How many neighbours?

Figure 4.1 sweeps k from 1 to 100. Two trends run in opposite directions, and both appear on both datasets:

- **Variant C improves with k and then plateaus**, flattening around $k \approx 20$ –50. Because C averages over the neighbourhood, its width is 87 columns whatever k is; larger k therefore only smooths sampling noise and costs nothing.
- **Variant D degrades monotonically with k** , and the decline tracks its column count exactly — 1740 columns at $k = 20$, 8700 at $k = 100$. It contains strictly more information than C and uses it strictly worse.

The conclusion is the clearest in this report, and it holds across two datasets, two model families and seven values of k : **a neighbourhood should be pooled, not enumerated**. $k = 20$ is used for all subsequent experiments. It is the onset of the plateau, and it is effectively free for variant C, whose column count does not depend on k (measured: 0.46 s to construct at $k = 20$ against 0.48 s at $k = 10$).

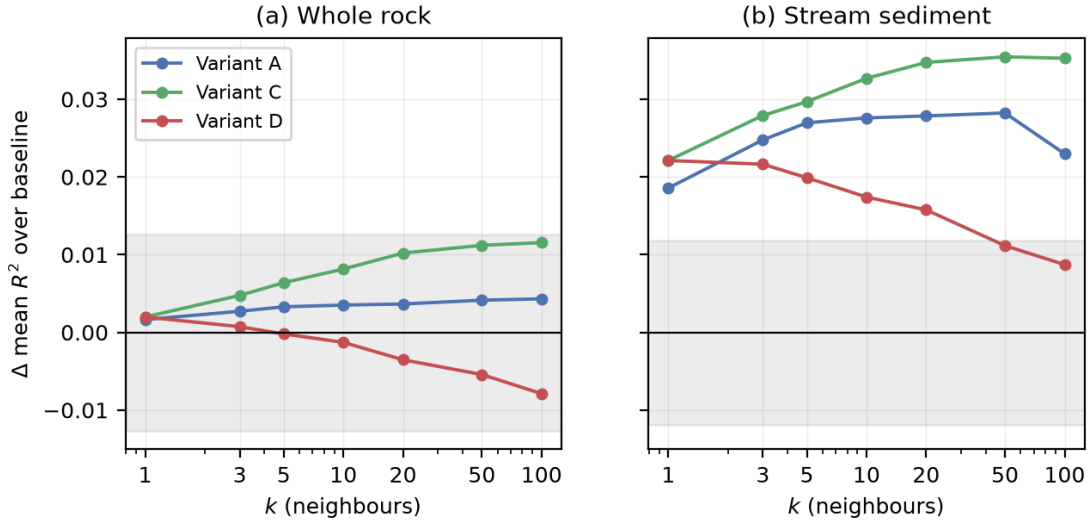


Figure 4.1: Change in mean R^2 over the no-neighbour baseline as the neighbourhood size k varies, for the ten best-covered elements of each dataset. The grey band is one fold standard deviation. Variant C (pooled) rises and plateaus; variant D (enumerated) falls in step with its column count. *The absolute levels here reflect the ten-element target sets and should be read alongside Section 4.5; the shape is what this figure establishes.*

4.5 A correction: how target selection manufactured an effect

An earlier version of this analysis scored both datasets on their ten best-covered elements. Under that choice, variant C was worth $+0.0102$ (0.81σ) on whole rock but $+0.0347$ (2.93σ) on stream sediment — a sharp contrast, which invited the reading that *sampling geometry* determines whether spatial context helps: dispersed sampling benefits, clustered exploration sampling does not.

That contrast does not survive the coverage threshold. Under the principled target sets, whole rock is essentially unchanged ($0.81\sigma \rightarrow 0.82\sigma$) while stream sediment collapses from 2.93σ to 0.91σ , leaving the two datasets statistically indistinguishable (Table 4.2).

The cause is a single element. Uranium’s neighbour gain on stream sediment is $+0.2205$, an order of magnitude larger than any other element’s. In a ten-element mean it contributes $+0.0220$ of the $+0.0342$ total — **65 % of the headline was one element** — while in a 71-element mean it contributes $+0.0031$ of $+0.0125$, or 25 %. Excluding uranium entirely, stream sediment’s gain is $+0.0096$.

The two top-ten sets happened to differ in a way that exaggerated the contrast from both ends: whole rock’s is dominated by major oxides that own chemistry predicts easily, stream sediment’s is trace-heavy and, critically, contained the one element where

neighbours genuinely pay. We report this at length because the failure is invisible from inside the original experiment: the numbers were correct, the folds were honest, and the conclusion was still an artefact of a convenience choice made before any modelling.

4.6 Where the gain lands

A pooled mean over dozens of elements is a weighted average of very different behaviours, which is exactly why it moved so much when the target set changed. Binning targets by *their own baseline score* separates elements that own chemistry already predicts from those it cannot (Figure 4.2 and table 4.3).

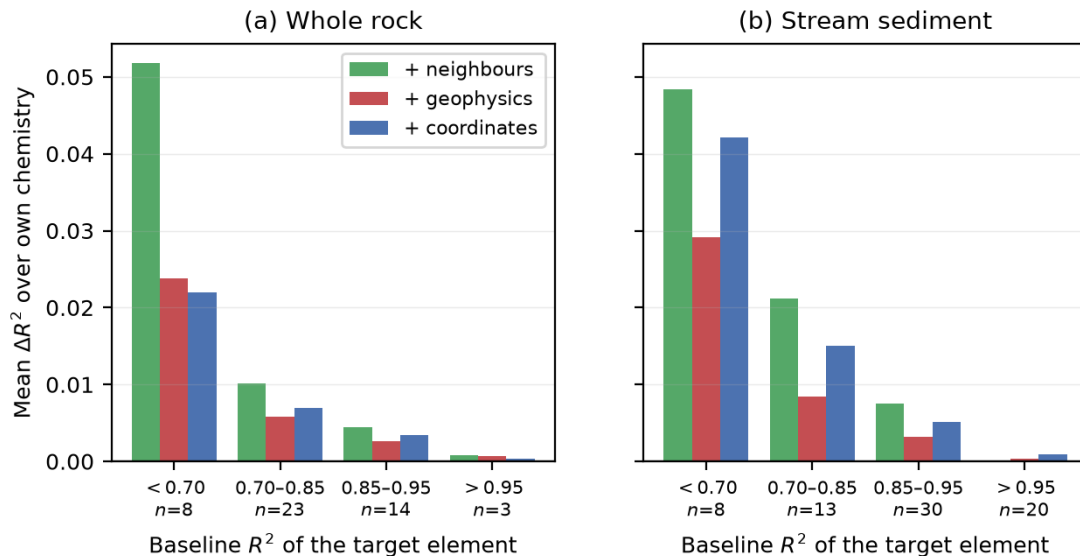


Figure 4.2: Mean improvement over own chemistry, with targets grouped by how well own chemistry already predicts them. Two datasets with different sampling geometries, different target sets and different element counts produce almost the same curve: spatial context is worth roughly +0.05 where chemistry fails and nothing at all where chemistry already succeeds.

The decay is monotone on both datasets and the two curves are nearly identical, despite the datasets differing in sample count, spatial extent, co-location rate and target composition. Where own chemistry already explains more than 95% of variance, spatial context contributes nothing measurable — which is unsurprising, since there is no headroom left to fill. Where own chemistry manages less than 0.70, it contributes around +0.05.

This, rather than any pooled mean, is the central result of RQ1. It also explains the correction of Section 4.5 mechanically: changing the target set changes the mixture of

Table 4.3: Mean gain over own chemistry by baseline difficulty band, $k = 20$, coverage-threshold target sets.

Baseline R^2	n	Mean baseline	+ neighbours	+ geophysics
<i>Whole rock</i>				
< 0.70	8	0.5920	+0.0517	+0.0238
0.70–0.85	23	0.7927	+0.0102	+0.0058
0.85–0.95	14	0.8909	+0.0044	+0.0026
> 0.95	3	0.9597	+0.0008	+0.0007
<i>Stream sediment</i>				
< 0.70	8	0.5846	+0.0484	+0.0291
0.70–0.85	13	0.7896	+0.0211	+0.0084
0.85–0.95	30	0.9076	+0.0075	+0.0032
> 0.95	20	0.9714	+0.0001	+0.0004

difficulty bands, and therefore the pooled average, without anything about the underlying behaviour changing at all.

4.7 The elements that benefit

The elements gaining most are not an arbitrary assortment, and they recur across the two datasets (Table 4.4). Arsenic, antimony, lead and caesium appear in both top-six lists, computed independently on different samples, different target sets and different sampling geometries.

Table 4.4: The six elements with the largest gain from neighbour features on each dataset, $k = 20$. Four elements (As, Sb, Pb, Cs) appear on both lists.

Whole rock		Stream sediment	
Element	ΔR^2	Element	ΔR^2
As	+0.096	U	+0.220
Au	+0.092	As	+0.060
Sb	+0.070	Br	+0.043
Cs	+0.048	Sb	+0.041
S	+0.040	Pb	+0.034
Pb	+0.027	Cs	+0.031

These are, with the exception of bromine, the classical hydrothermal pathfinder suite together with mobile alkalis — elements whose distribution is governed by regional-scale processes such as mineralisation and alteration rather than by bulk rock composition.

Uranium is the extreme case: on stream sediment it rises from 0.594 on own chemistry alone to 0.814 with neighbours. Uranium is also the only element with a reliable geophysical correlation in this data (+0.28 with magnetic anomaly and -0.34 with Moho depth, $n = 3227$), which is coherent, since uranium concentrates in evolved felsic crust that is thick and weakly magnetic.

An important caveat is recorded in Section 5.5: these elements are also among the hardest targets, and Section 4.6 shows gain tracks difficulty. The geological and the statistical explanation are confounded in the present design.

4.8 Where the spatial signal comes from

Figure 4.3 and table 4.5 score each feature source alone and in combination. Three readings follow.

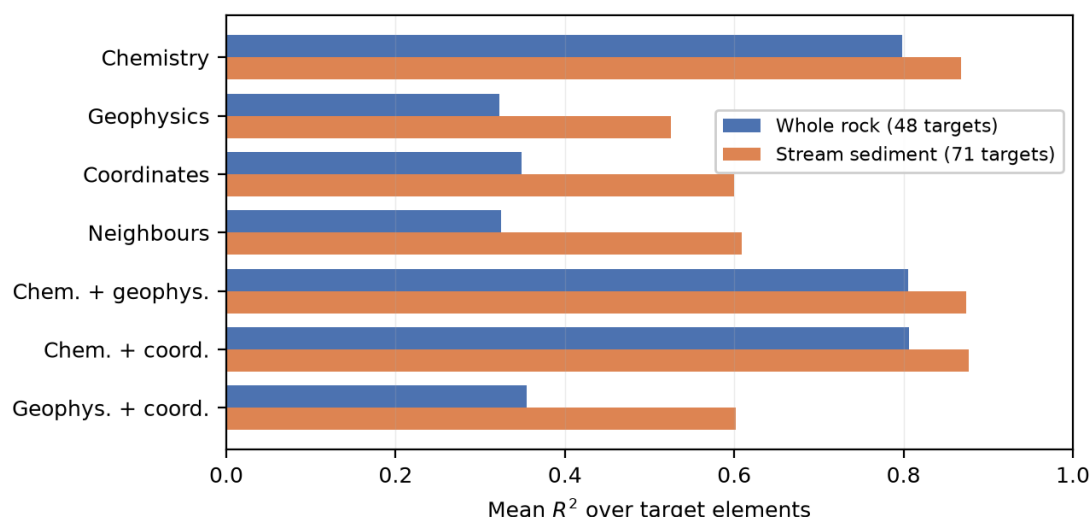


Figure 4.3: Mean R^2 by feature source on both datasets. Standalone spatial sources are roughly twice as informative on stream sediment as on whole rock, but every one of them adds about the same small amount on top of own chemistry.

Sampling geometry decides how much spatial information exists. Every standalone spatial source is roughly twice as informative on stream sediment as on whole rock: neighbours 0.3246 against 0.6086, coordinates 0.3484 against 0.5994. This is a large, robust difference, and it is the part of the original sampling-geometry hypothesis that survives.

Table 4.5: Feature-source ablation, coverage-threshold target sets. The neighbour row uses variant C at $k = 20$.

Feature set	Whole rock	Stream sediment
Chemistry	0.7983	0.8676
Geophysics	0.3228	0.5253
Coordinates	0.3484	0.5994
Neighbours	0.3246	0.6086
Chemistry + geophysics	0.8059	0.8739
Chemistry + coordinates	0.8063	0.8775
Geophysics + coordinates	0.3546	0.6017

But it does not decide how much that information adds. Five times the spatial signal buys the same sub- σ improvement on top of own chemistry. The reason is redundancy: a sample’s own chemistry already encodes most of what its location would tell you, because chemistry is what location predicts.

The three spatial sources are interchangeable. Geophysics, coordinates and neighbours land in one narrow band on each dataset, and each adds between +0.006 and +0.012 on top of chemistry. In particular, **the geophysical grids do not beat plain coordinates**: coordinates outscore geophysics on both datasets, and geophysics combined with coordinates barely exceeds coordinates alone (0.6017 against 0.5994 on stream sediment). On this evidence the grids encode *where* a sample is rather than independently *what* is underfoot. Section 5.5 explains why this particular reading is the one most likely to change under spatial cross-validation.

4.9 Does a learned aggregation do better?

Figure 4.4 compares both graph architectures against variant C across the same seven values of k .

The hand-built encoding wins everywhere. Variant C is above both networks at all seven values of k , with no crossover. The best network configuration is GRAPH-SAGE at $k = 20$ (0.8260); the best hand-built result is variant C at $k = 50$ (0.8342).

Distance-aware attention does not pay, and eventually hurts. The two architectures cross at $k \approx 5$. Below it the attention model is better, as designed — with few neighbours, distance discriminates usefully. Above it the distance-blind control wins by a margin that widens monotonically to +0.0140 at $k = 100$. Run-to-run variation between repeats of

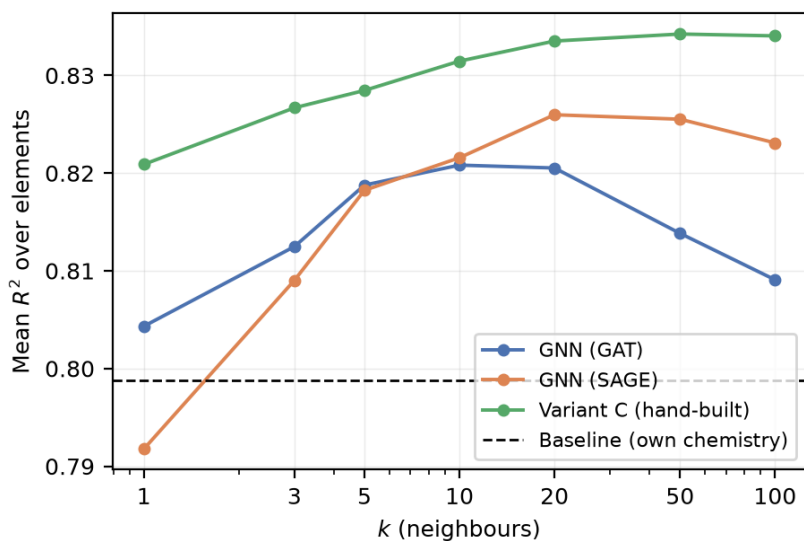


Figure 4.4: Mean R^2 on stream sediment for the two graph architectures and the hand-built pooled encoding, across neighbourhood size. Variant C dominates both networks at every k . The distance-aware attention model (GAT) beats the distance-blind control (SAGE) only below $k \approx 5$, and degrades above $k = 20$.

the same configuration is approximately 0.002, so the gaps at $k \geq 20$ are well outside it. The attention model also degrades past $k = 20$ (0.8205 to 0.8091), tracking edge count in exactly the way variant D tracks column count.

On the widened target set the network matches the baseline, not the encoding.

Running the settled configuration (GRAPHSAGE, $k = 20$) on all 71 stream-sediment targets gives 0.8676 — identical to the no-neighbour baseline, against variant C’s 0.8801. One target distorts this: `Sum wt%` is a -0.253 outlier for the network where nothing else is worse than -0.061 (see Section 5.5). Excluding it, the network is worth $+0.0037$ against variant C’s $+0.0128$, so it recovers roughly 29 % of the available gain rather than none. Adding geophysical or coordinate channels to the node features moves the mean by $+0.0004$ and $+0.0008$ respectively — inside noise. On uranium the network exactly ties the hand-built columns (0.814 against 0.814): it captures the single large spatial signal and misses the many small ones.

4.10 Model independence

The central claim of this chapter is a null result, so it was checked under a second inductive bias. On whole rock, all four encodings were run under both a random forest

and gradient boosting, with the same conclusion under each: every gain inside one fold standard deviation, and variant D worst (Table 4.6).

Table 4.6: Change over the no-neighbour baseline under two model families, whole rock, ten elements, co-located neighbours excluded.

Variant	Random forest	Gradient boosting
A	+0.0016	+0.0033
B	−0.0004	+0.0033
C	−0.0003	+0.0064
D	−0.0101	−0.0002

Stream-sediment results use gradient boosting only. Repeating the full A–D comparison under a random forest there was considered and rejected: variant D at $k = 20$ is 1740 columns and `RandomForestRegressor` evaluates all features at every split, making it far more expensive than the histogram method for a confirmation the whole-rock table already provides. This is a stated limitation rather than an oversight.

5 Discussion

5.1 Spatial context is redundant, not absent

The single sentence that summarises RQ1 is that *a sample’s own chemistry already contains most of what its location could tell you*. Every route we tried into the spatial signal — averaging neighbouring samples, three independent geophysical fields, raw projected coordinates, and a learned graph aggregation — lands in the same narrow band of improvement, between +0.006 and +0.012 mean R^2 , all of it under one fold standard deviation.

That several very different encodings converge on the same small number is itself the argument. If the limit were an encoding problem, one of them should have done materially better. The consistent ceiling instead suggests the limit is informational: after conditioning on 48 or 71 co-measured elements, there is little left for position to explain.

This is not the same as saying spatial structure is absent. Section 4.8 shows a large amount of it — neighbours alone reach R^2 0.6086 on stream sediment with no access to the sample’s own chemistry whatsoever. The signal is real, substantial, and almost entirely redundant with information the analyst already has.

5.2 Why pooling beats enumeration

Variants C and D contain the same measurements. C averages each element over the neighbourhood and emits m columns; D lists every element of every neighbour and emits mk . C improves with k ; D degrades with k , and the degradation tracks the column count on both datasets and in the graph network’s edge count as well.

The interpretation is that a neighbourhood carries a small amount of signal spread thinly across many correlated measurements. Averaging concentrates it and suppresses per-sample noise at fixed model capacity. Enumerating preserves the same signal but dilutes it across a feature space the learner must now search, and the search cost exceeds the informational gain. Notably, the neighbour ordering that variant D preserves — which sample was nearest — turns out not to be worth its dimensionality, which is consistent with the graph network’s finding that edge distance adds nothing once the neighbourhood is reasonably large.

5.3 Why the graph network does not win

The network was given every structural advantage: the same folds, the same neighbourhood, a skip connection so the graph only had to add what it knew, and an architecture able to read edge distance. It still does not beat an average.

Three non-exclusive explanations, in the order we find them plausible. First, with only element concentrations as node features, the messages passed between neighbours carry little that a per-element neighbour mean does not already summarise — and gradient boosting is simply a stronger tabular learner at this data scale [5]. Second, the graph is purely geometric: a k -nearest-neighbour graph in metres encodes proximity, whereas the geologically meaningful adjacency is shared catchment, shared lithology or shared intrusive suite, none of which is available to it. Third, the network is untuned beyond one hidden width and one learning rate, so some margin certainly remains — though not, we would argue, the +0.0125 needed to overtake a 87-column average.

5.4 The methodological result

The finding we expect to be most useful to follow-up work on this archive is not about neighbours at all. Scoring on the ten best-covered elements — an entirely reasonable-looking convenience — produced a clean, interpretable, and false conclusion about sampling geometry that survived model comparison, a k -sweep and a co-location control, because none of those checks touch target selection. It was only visible after replacing the cut with a coverage threshold.

The general lesson is that in a multi-target problem, the choice of which targets to score is a modelling decision with the same status as the choice of features or folds, and it should be justified rather than defaulted. It is worth adding that the artefact was not random: the convenience cut favoured exactly the elements with the most data, which on stream sediment included the one element where the effect under study is genuinely large.

5.5 Limitations

Random cross-validation. All results use random folds. Position is unusually informative in this setting, because test samples frequently have training samples within a few hundred metres. The reading most exposed to this is Section 4.8’s conclusion that geophysics does not beat coordinates: under spatial blocking, coordinates should degrade sharply — a model that has learned “this northing implies this uranium level” is useless when that whole band is held out — while the geophysical grids remain real measurements available at every point. That comparison could invert, and testing it is the first task of RQ2.

Two derived target columns. Sum wt% in the stream-sediment dataset is exactly the sum of the major oxides plus loss on ignition (median absolute difference 0.0000 % over 8482 samples, with 98.9 % within 0.5 %). The relation runs in both directions, so the major oxides are inflated by subtraction as well. Separately, about ten elements appear twice under different unit conventions (Fe as Fe % and Fe2O3 wt%, $r = 0.93$; Mn likewise, $r = 0.97$), which is redundancy rather than leakage since the pairs come from different digestions. Both effects raise absolute baselines and dilute measured gains. Neither affects any conclusion here, because every claim is a *difference* between two arms that see identical features — but the > 0.95 band of Table 4.3 is partly populated by these columns and should not be over-interpreted. The < 0.70 band, which carries the finding, is unaffected.

Confounded explanation of the pathfinder result. The elements that gain most are mobile, hydrothermally mobilised elements — and they are also the elements own chemistry predicts worst. Section 4.6 shows gain tracks difficulty. Distinguishing “spatial context helps mobile elements” from “spatial context helps hard targets” would require comparing elements matched on baseline score, which this design does not do.

Scope of the model check. The two-model comparison covers whole rock only (Section 4.10). Five stream-sediment targets have fold standard deviations above 0.03 (Ta, Au, U, Pr, Cl) and gold is effectively unpredictable from chemistry (R^2 0.128). The graph network was tuned only lightly, and its hyperparameters were not searched.

6 Conclusion and future work

We built one preprocessing and evaluation pipeline and applied it unchanged to two independent GEUS geochemistry datasets in order to answer RQ1: whether a held-out element can be predicted from a sample’s other elements, and whether spatial context helps.

It can, and it largely does not. Own chemistry reaches a mean R^2 of 0.868 on stream sediment and 0.798 on whole rock across 71 and 48 target elements. Neighbouring chemistry, geophysical grids and raw coordinates each add less than one fold standard deviation on top, they are interchangeable with one another, and a graph neural network with learned distance-weighted aggregation does not beat a plain average over the same neighbourhood. Where the residual gain does land is systematic: roughly +0.05 for elements own chemistry predicts poorly, nothing for elements it already predicts well, with the two datasets tracing almost the same curve, and concentrated in a coherent set of pathfinder and mobile elements led by uranium. Along the way we found that a conventional convenience — scoring the ten best-covered elements — had manufactured an apparent dataset-level effect of which 65 % was a single element.

6.1 Future work: RQ2

The natural continuation, and the one the supervisor’s framing asks for, is to hold out *regions* rather than samples:

1. **Spatial block folds.** Hash projected coordinates into squares of side `blockKm` and assign whole blocks to folds, written in the same format as the existing fold files so no downstream code changes. Blocks are preferable to clustering, because block size in kilometres is an interpretable parameter whereas clusters follow sampling density [4].
2. **Re-run the core comparison.** Baseline, variant C and the full feature-source ablation. This is the step that can change a conclusion: if geophysics overtakes coordinates under spatial hold-out, that is a substantive result and precisely the geologically motivated angle the project set out to add.
3. **A block-size sweep.** Reporting R^2 against block size gives a degradation curve — the distance over which the model generalises — which is a more useful deliverable to a geologist than any single number.

4. **Geological province hold-out.** The stricter test: a coherent terrane rather than an arbitrary square, using the published subglacial province polygons.

RQ2 will use stream sediment only. Whole rock spans 11.7° of longitude within a single region of West Greenland, so there is no meaningful “outside” to hold out, and block cross-validation would either produce blocks too small to be spatially independent or fragment the dataset. Its role in this project was as the control that made the sampling-geometry comparison possible, and that role is complete.

Beyond RQ2, the most promising direction suggested by Section 4.9 is not a better network but a better graph. The present adjacency is purely geometric; encoding shared catchment, mapped lithology or intrusive suite would give message passing something a distance matrix cannot express, which is the condition under which a graph model could be expected to overtake a well-chosen average.

Declaration on the use of AI tools

This project used Claude (Anthropic) via the Claude Code environment [16] as a programming assistant for the pipeline, experiments and figures, and in drafting this report from the experimental outputs. All reported numbers are produced by running the included notebooks against the committed data and are reproducible from the repository. I have reviewed and verified the content and take full responsibility for it.

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A Additional Material

A.1 Reproduction

Three notebooks, run in order:

1. `Data Fetching.ipynb` — preprocessing, geophysical attachment, splits.
2. `Random Forest.ipynb` — tabular models, neighbour encodings, k -sweep, feature-source ablation.
3. `GNN.ipynb` — graph construction, both architectures, k -sweep, settled configuration.

Switching dataset requires changing three constants (`DATA_TYPE`, `MERGE_TYPE`, `SOURCE_CSV`) and nothing else. Every experiment checks a JSON cache in `Results/` before training, so a re-run is near-instant unless a file is missing. Random seeds are fixed at 42 throughout. The graph experiments require a GPU runtime; all tabular experiments run on CPU.

A.2 Fold noise floor

The mean across-fold standard deviation of R^2 , averaged over target elements, is 0.0180 on whole rock and 0.0137 on stream sediment. Differences below this are not interpreted as real anywhere in this report.